

NIORT SOUCHE AP, France

WMO: 073300

Lat: 46.3156N

Lon: 0.4003W

Elev: 62

StdP: 100.58

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.1	-2.4	-7.3	2.0	-2.6	-5.3	2.4	-0.9	13.4	11.4	11.9	10.0	2.9	60	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.1	20.6	30.0	19.7	28.0	19.1	21.8	29.1	20.9	27.8	20.0	26.2	3.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.4	24.5	18.6	13.6	23.1	17.8	12.9	22.1	64.0	29.3	60.5	27.9	57.4	26.2	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	9.0	8.0	DB	-6.2	35.9	2.3	1.7	-7.9	37.1	-9.3	38.0	-10.6	39.0	-12.3	40.2	(4)
			WB	-6.7	23.5	2.3	1.0	-8.4	24.2	-9.7	24.8	-11.0	25.3	-12.7	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	5.6	6.3	8.9	11.2	14.8	18.5	20.3	20.1	17.2	13.9	8.9	6.2
	DBStd	6.30	3.83	3.62	3.26	3.27	3.22	3.30	3.12	3.12	3.03	3.46	3.54	3.95
	HDD10.0	527	140	108	60	23	2	0	0	0	0	8	60	126
	HDD18.3	2277	394	336	293	215	119	37	14	14	57	140	282	376
	CDD10.0	1514	4	5	25	58	149	255	321	314	216	131	28	8
	CDD18.3	223	0	0	0	1	8	42	76	69	23	4	0	0
	CDH23.3	2171	0	0	1	18	99	444	708	669	212	19	0	0
	CDH26.7	738	0	0	0	1	19	150	253	259	54	2	0	0
Wind	WSAvg	3.8	4.2	4.3	4.2	3.9	3.6	3.5	3.5	3.2	3.4	3.5	3.8	4.1
Precipitation	PrecAvg	887	91	74	61	71	68	55	55	66	93	99	100	
	PrecMax	1185	166	146	202	222	135	122	142	137	231	183	240	193
	PrecMin	556	17	6	3	13	18	12	4	5	12	12	25	18
	PrecStd	145	44	42	46	45	31	26	27	38	49	46	50	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	17.1	21.4	25.6	29.2	33.6	34.7	35.7	30.9	25.6	18.9	14.8
		MCWB	12.3	11.6	13.6	16.4	19.1	21.4	20.5	21.6	19.5	18.0	14.9	12.7
	2%	DB	13.0	14.2	18.7	22.7	26.1	30.4	31.7	32.1	28.1	22.9	16.7	13.7
		MCWB	11.7	10.6	12.4	15.1	17.6	20.1	20.2	21.0	18.4	17.0	14.2	12.3
	5%	DB	12.0	12.5	16.3	19.9	23.7	28.1	29.5	29.6	25.7	20.8	15.2	12.7
		MCWB	10.8	10.2	11.6	13.7	16.6	19.4	19.5	20.1	17.7	16.5	13.4	11.5
	10%	DB	11.0	11.4	14.2	17.5	21.4	25.6	27.3	27.0	23.6	19.0	14.0	11.8
		MCWB	9.9	9.7	10.7	12.4	15.7	18.3	18.8	19.0	16.8	15.8	12.5	10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	12.6	14.4	17.3	20.5	22.9	22.5	23.7	20.9	19.2	15.9	13.6
		MCDB	13.5	14.7	19.5	24.0	27.2	30.1	30.4	32.1	26.6	22.7	17.2	14.2
	2%	WB	12.0	11.5	13.2	15.6	18.6	21.3	21.2	22.1	19.6	18.1	14.7	12.6
		MCDB	12.7	13.2	17.1	21.5	24.1	28.4	28.9	29.0	25.3	21.0	16.1	13.4
	5%	WB	11.0	10.8	12.3	14.2	17.2	20.2	20.4	20.9	18.6	17.1	13.7	11.7
		MCDB	11.9	12.2	15.2	18.9	22.4	26.6	27.3	27.8	23.7	20.0	15.0	12.5
	10%	WB	10.0	9.9	11.3	12.9	16.1	18.9	19.5	19.8	17.7	16.2	12.6	10.8
		MCDB	10.9	11.2	13.6	16.6	20.5	24.5	25.5	25.5	22.1	18.6	13.9	11.7
Mean Daily Temperature Range	5% DB	MDBR	5.9	7.5	9.1	10.1	10.2	10.9	11.0	11.3	10.8	8.6	6.8	6.1
		MCDBR	6.0	9.0	12.4	13.9	13.9	14.8	14.4	15.5	14.1	10.6	7.4	6.1
		MCWBR	5.0	6.3	7.4	7.7	7.0	6.3	5.5	6.2	6.2	5.8	5.3	5.0
	5% WB	MCDBR	5.4	7.8	10.0	12.7	12.4	13.4	12.3	13.4	11.8	8.9	6.9	5.7
		MCWBR	4.9	5.8	6.4	7.3	6.9	6.4	5.4	5.9	6.0	5.3	5.4	5.1
Clear-Sky Solar Irradiance	taub	0.325	0.341	0.371	0.395	0.399	0.412	0.402	0.392	0.366	0.364	0.338	0.321	
	taud	2.444	2.389	2.307	2.240	2.282	2.280	2.305	2.352	2.401	2.418	2.436	2.433	
	Ebn at Noon	768	827	850	860	868	855	859	854	847	789	742	727	
	Edh at Noon	70	89	111	129	129	130	125	115	100	85	70	64	
All-Sky Solar Radiation	RadAvg	1.20	2.14	3.32	4.71	5.53	6.22	6.01	5.32	4.21	2.52	1.45	1.05	
	RadStd	0.12	0.22	0.37	0.47	0.43	0.50	0.30	0.40	0.27	0.21	0.15	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page